

ExamGadh Architecture & Design

DRAFT — pending client approval

Version 0.1 · 27 Oct 2026
Based on requirements v0.3
CGPSC & Vyapam exam-prep platform: web, Android, admin panel, backend

ExamGadh – Architecture & Design

DRAFT — pending client approval. Step 2 of the SDLC pipeline. Build (step 3) stays locked until the client approves this document.

1. Summary and decisions

ExamGadh is a Hindi/English exam-prep platform for CGPSC and CG Vyapam aspirants: tests (static and live), notes, PYQs, current affairs, analytics, doubts, 1v1 battle, study planner and interview prep, sold by subscription through Razorpay. The first-year scale is small (2k monthly active users, 300 concurrent on a live test), the budget is Rs 5,000–15,000/month, and the team is two people. The design is therefore deliberately plain: **one TypeScript backend, one PostgreSQL database, one small server per environment**, no Kubernetes, no microservices.

#	Decision	Choice	Alternative rejected	Reason
D1	Cloud / region	AWS Mumbai (ap-south-1)	Other cloud; other region	India data residency and latency for CG users; user expects AWS (pipeline default)
D2	Backend	NestJS (Node 22, TypeScript) modular monolith, REST + WebSocket	Python/Django; microservices	One language across web, mobile and backend (shared types); WebSocket for live test/battle in the same process; 2-person team
D3	Database	PostgreSQL 16 on RDS	DynamoDB; MongoDB	Relational content (exam→subject→topic→question→test), joins for analytics, JSON columns for bilingual question bodies; see §5.3
D4	Cache / realtime state	Redis 7 in a container on the app server	ElastiCache	ElastiCache costs ~Rs 1,100+/month extra at the smallest size; Redis here holds only rebuildable data (sessions index, battle queue, leaderboards, rate limits)
D5	Compute	One EC2 t4g instance per environment (Docker Compose: api, web, admin, redis, caddy), auto-recovery enabled	ECS Fargate + ALB; Kubernetes	ALB alone is ~Rs 1,800/month; at 300 peak concurrent a single instance is enough; simple to run for 2 people
D6	Web	Next.js (student app + a separate admin app)	One Next.js app with /admin	Admin is a different trust boundary: own hostname, 2FA, optional IP allow-list
D7	Mobile	React Native with Expo (development build / EAS)	Bare RN; Flutter	Razorpay's RN SDK needs native code, which Expo dev builds support via a config plugin; code sharing with web

#	Decision	Choice	Alternative rejected	Reason
D8	Auth	Google Sign-In (OIDC ID token verified server-side) + own short-lived JWT + server-side session record. **No phone number at launch** (user decision)	Phone OTP; Firebase Auth	Single-active-session rule needs sessions the server controls; no SMS cost
D9	Payments	Razorpay only (Orders API, Checkout web / RN SDK, signed webhooks)	Play Billing; other gateways	User decision. Risk: Play policy (see §9)
D10	Files	S3 + CloudFront (notes images, PYQ PDFs, doubt photo/voice, CA cards)	Store in DB	Cheap, cacheable; pre-signed uploads
D11	IaC / CI	Terraform; GitHub Actions with OIDC to AWS (no long-lived keys)	Console clicks; CDK	Repeatable `dev` now, `prod` later from the same modules
D12	Environments	**dev now, prod later** (user decision); no always-on staging	dev+staging+prod	Saves cost during development. Dev doubles as the pre-prod test bed until prod exists
D13	Push	Firebase Cloud Messaging	SNS mobile push	Free, standard for RN
D14	Background jobs	BullMQ on the same Redis	SQS + Lambda	One less AWS service; jobs are light (notifications, SLA timers, result ranking, reports)

2. Sizing inputs and assumptions

Answers from the sizing interview (2026-10-27):

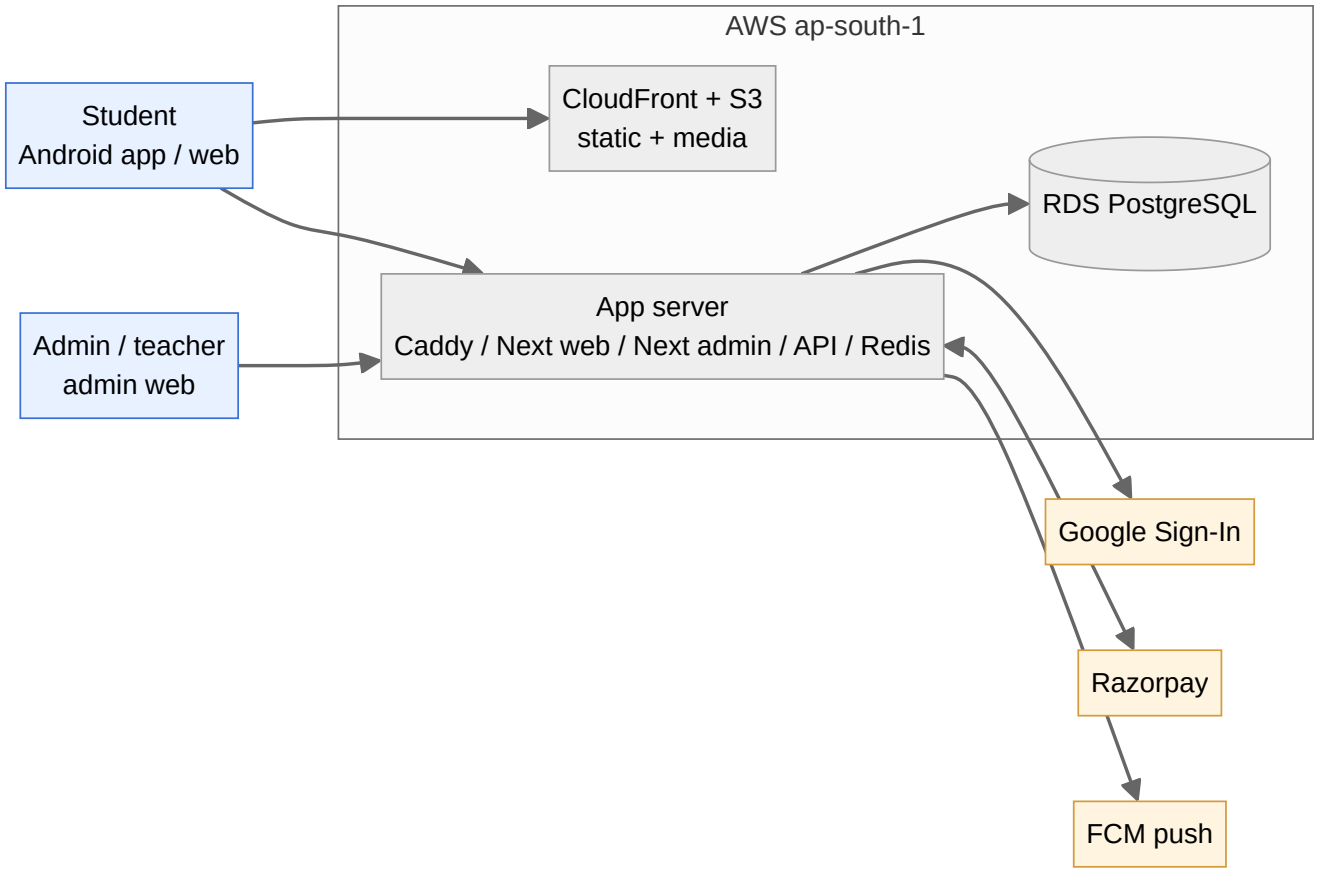
Input	Value	Source
Registered users, 12 months	5,000	user (option "Small")
Monthly active users	2,000	user
Peak concurrent (live test)	300	user
Monthly infra budget	Rs 5,000–15,000 (target, not hard cap)	user
Availability target	99.5% (~3.6 h/month), no on-call rota, 2-person team	user

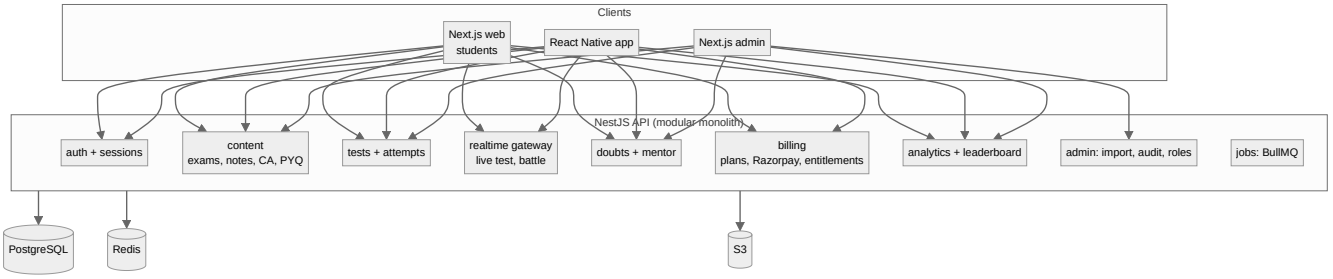
Input	Value	Source
Region	Mumbai	assumed (not objected to)
Environments	dev first, create prod later	user
Phone number	none at launch	user

Assumptions (derived, to be validated at load test, step 5):

- A live test of 100 questions with 300 users: each answer sync is a small write; ≈ 300 × (1 write / 20 s) ≈ 15 writes/s peak, plus the start/submit spike (≈ 300 requests in the first 10 s). One 4-GB instance handles this comfortably.
- Question data: 20,000 questions × ~2 KB bilingual ≈ 40 MB. Notes and PYQ PDFs ≈ 2 GB in S3. Attempts: 2,000 MAU × 20 tests × 100 answers = 4 M `attempt_answers` rows/year ≈ 1–2 GB. Everything fits in a 20–50 GB database.
- Reads:writes ≈ 90:10 outside live tests; content is read-heavy and cacheable (CloudFront, Redis).
- Peak days (exam week, results) may be 3× normal; instance can be resized in minutes.
- Retention: attempts kept for the life of the account; audit log 12 months; backups 7 days (dev), 14 days (prod).
- Growth beyond 10k MAU or 1,000 concurrent: split web from API onto a second instance, add read replica; no redesign needed.

3. System context and components

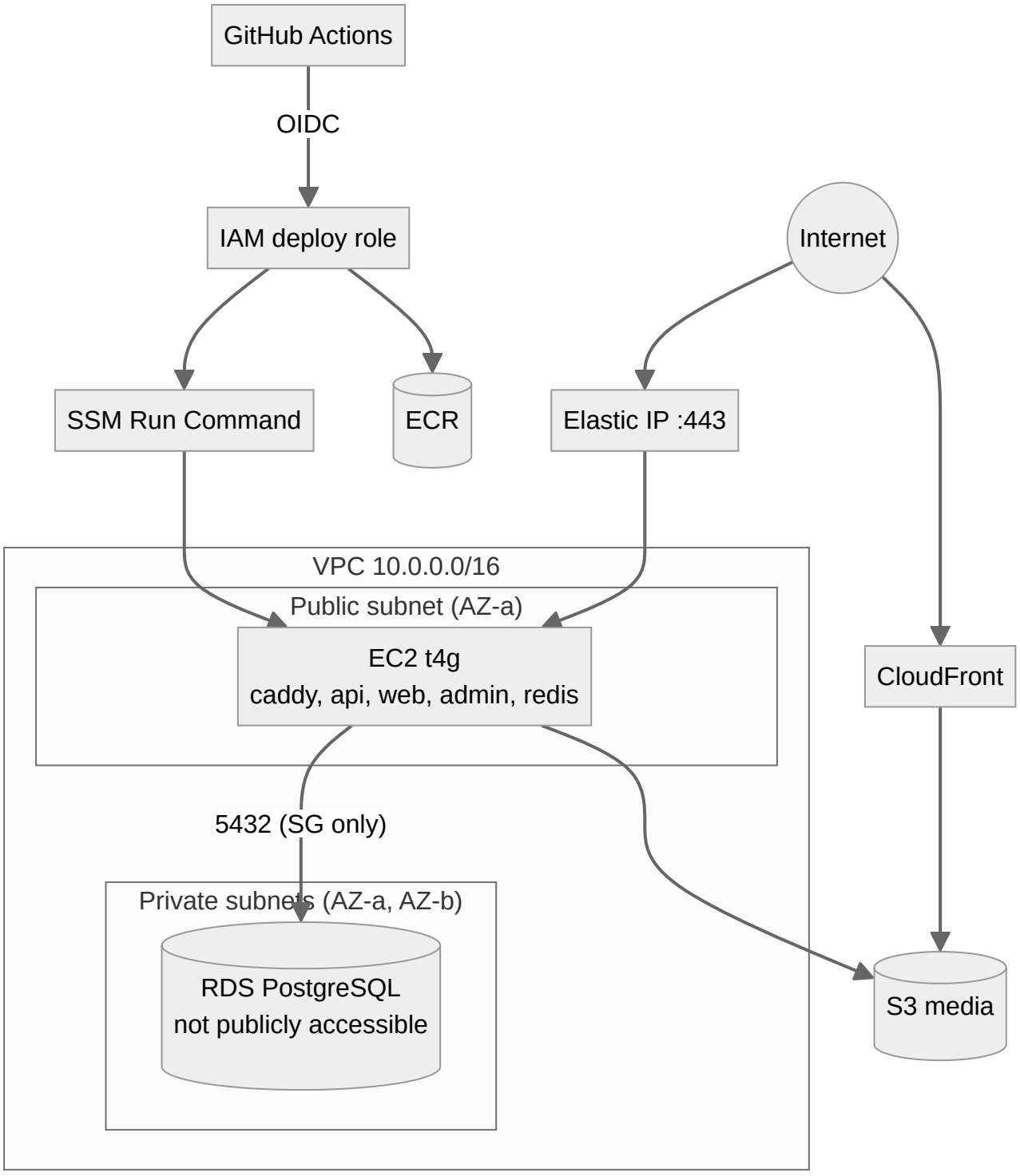




Module boundaries are folders with explicit interfaces inside one deployable. Splitting later (for example, realtime) is possible but not planned.

4. Infrastructure (AWS)

- **Network:** one VPC, 2 AZs defined (needed by RDS subnet groups). Public subnet: the app instance (Elastic IP, security group allows 80/443 from CloudFront ranges and the world for the API host). Private subnets: RDS (no public access, only the app SG on 5432).
- **Compute:** EC2 `t4g.medium` (2 vCPU, 4 GB, ARM) in prod; `t4g.small` in dev, stopped nights/weekends by a schedule. Docker Compose runs `caddy` (TLS via Let's Encrypt), `api`, `web`, `admin`, `redis`. Instance recovery alarm restarts on hardware failure. Deploy via GitHub Actions → ECR → SSM Run Command (no SSH).
- **Database:** RDS PostgreSQL `db.t4g.small` prod / `db.t4g.micro` dev, gp3 20 GB with autoscaling to 100 GB, automated backups (14 / 7 days), encryption at rest with KMS. Single-AZ (99.5% target); Multi-AZ is the first upgrade if the SLA must rise.
- **Storage / CDN:** S3 buckets (`media`, `backups`, `terraform state`) with CloudFront for `media`; public bucket access blocked, CloudFront OAC.
- **DNS / TLS:** Route 53 (or existing registrar DNS); hostnames `app.`, `admin.`, `api.`.
- **Secrets:** AWS Secrets Manager / SSM Parameter Store (Razorpay keys, Google client id, DB URL, JWT keys). Parameter Store standard parameters are free; Secrets Manager only for DB credentials rotation.
- **Observability:** CloudWatch Logs + metric alarms (CPU, disk, 5xx, RDS free storage/connections), uptime check, SNS → email/WhatsApp-able channel. Sentry (free tier) for app errors.
- **Cost controls:** AWS Budgets alert at 60/80/100% of the monthly cap; dev schedule (stop EC2 and RDS out of hours); S3 lifecycle rules.



5. Backend, frontends, data

5.1 Backend (repo `exam-gadh-backend`)

- NestJS + TypeScript, Prisma ORM, class-validator, OpenAPI generated from code. REST for CRUD and a Socket.IO gateway for live tests and battles.
- **Auth and single-active-session (FR-001, FR-003):**
  1. Client sends Google ID token; server verifies signature and `aud`, finds/creates the user.
  2. Server creates a `session` row (user, device id, platform, created/last_seen) and **revokes every other active session of that user** (newest wins; web and app count as one).

3. Access JWT lives 15 min and carries the `sid`; refresh token is rotated and stored hashed. Each request, and each WebSocket heartbeat (30 s), checks `sid` against a Redis set of valid sessions; a revoked session gets `401 SESSION_REPLACED` and the old device shows "signed in elsewhere".
4. Admin dashboard shows concurrent sessions per user and flags accounts that flip between devices repeatedly (sharing detector).

- **Tests and attempts:** server is the time authority (`/time` offset, attempt `deadline_at`); answers saved incrementally (autosave) so a dropped network does not lose work; submit is idempotent; scoring with negative marking runs on the server; results and rank computed by a job at test close for live tests.
- **Question bank (FR-013):** questions are tagged exam/subject/topic/subtopic/difficulty/source; "pick" (manual) and "autopick" (random with constraints) build tests; personalised picks use per-user topic accuracy.
- **Live test / pre-test (FR-017):** scheduled start; lobby opens 10 min before; all clients start from a server `start_at`; late entry closes at `entry_close_min`.
- **Battle (FR-050/051):** 7-question matchups through a Redis queue; if no human is found within 8 s, the opponent is a replayed earlier attempt (answers + timings of a real prior user) or a bot with accuracy near the player's level. Results are written after both sides finish.
- **Payments (FR-081):** `POST /billing/orders` creates a Razorpay order for a plan; client pays via Checkout (web) or the RN SDK; client posts `razorpay_payment_id/order_id/signature` to `/billing/verify`, and the Razorpay `payment.captured` webhook (HMAC-verified, idempotent) is the source of truth that grants the entitlement. Refunds via admin. Coupons apply at order creation.
- **Doubts (FR-060/061):** quotas, 48 h SLA timers (job), similar-question FAQ search using Postgres full-text (Hindi/English) – OpenSearch is not needed at this size.
- **Analytics (FR-040/042):** nightly rollups into summary tables (per user-topic accuracy, per test statistics); percentile from stored distributions.
- **Admin import (FR-090):** template download (.xlsx/CSV) → upload → server-side validation with row-level errors and bilingual checks → preview → commit in one transaction with versioning; question lifecycle draft → approved → published with `publish_after`.
- **Notifications (FR-084):** FCM tokens per session; quiet hours (22:00–07:00) and a 2/day cap enforced by the sender job.

5.2 Frontends (repo `exam-gadh-frontend`, pnpm monorepo)

- `apps/web` – Next.js (App Router), Hindi-first i18n (next-intl), SSR for public pages (exam landing, sample questions, notes previews) for SEO; client-side test player.
- `apps/admin` – Next.js, separate hostname, TOTP 2FA, role-based UI.
- `apps/mobile` – React Native (Expo dev build), shared API client and types from `packages/api` (generated from OpenAPI), offline cache of notes and the in-progress attempt (MMKV/SQLite), FCM, Razorpay RN SDK.
- `packages/ui` (tokens, components), `packages/i18n` (Hindi and English strings), `packages/api`.
- Fonts: Noto Sans Devanagari bundled; no external font fetch in the test player.

5.3 Database choice

Chosen: PostgreSQL. The domain is relational (exam → subject → topic → question → test → attempt → answer, plus users, plans, entitlements, payments) and needs ranking, per-topic aggregates and transactional writes (payment + entitlement, import commits). JSONB holds

bilingual question bodies (`{en:{q,o[],s}, hi:{...}}`) without a schema per language; GIN full-text covers doubt similarity. **Alternative rejected: DynamoDB.** Cheap and elastic, but analytics, admin filtering, and ad-hoc reports (the admin app calls ~40 different list endpoints) would each need a purpose-built index; no joins; the team would pay in complexity at a scale where RDS costs ~Rs 2,500.

Outline of main tables (names are provisional):

Area	Tables
Identity	`users` (google_sub, email, name, avatar, district, target_exam, phone NULL), `sessions`, `devices`, `admin_users`, `admin_roles`, `audit_log`
Taxonomy	`exams`, `exam_papers`, `subjects`, `topics`, `syllabi`, `test_sections`
Content	`notes`, `note_sections`, `questions`, `question_versions`, `passages`, `pyq_papers`, `ca_items`, `ca_categories`, `faqs`, `updates`, `calendar`, `interview_questions`
Tests	`tests`, `test_section_items`, `test_questions`, `attempts`, `attempt_answers`, `battles`, `battle_rounds`, `leaderboards` (materialised)
Commerce	`plans`, `plan_prices`, `entitlements`, `orders`, `payments`, `coupons`, `access_rules`
Support	`doubts`, `doubt_messages`, `canned_replies`, `error_reports`, `mentor_requests`, `mentor_plans`, `notifications`, `fcm_tokens`
Ops	`import_jobs`, `settings`, `backups_log`

6. CI/CD and environments

- Repos: `exam-gadh-frontend`, `exam-gadh-backend`, `exam-gadh-terraform` (see amendment A1), `exam-gadh-docs`.
- Workflows (GitHub Actions):
  - **PR:** lint, typecheck, unit tests, `terraform fmt/validate`, dependency and secret scan.
  - **Merge to `main` → dev:** build images (arm64), push to ECR, run DB migrations, deploy via SSM, smoke test.
  - **Prod (when created):** same artefact promoted by a manual-approval job (GitHub Environment `prod` with required reviewer).
  - **Mobile:** EAS build (internal APK) on tag for dev; Play upload (or direct APK link) for prod.
- AWS auth: GitHub OIDC → IAM role scoped per repo and environment. No long-lived keys.
- Environments: `dev` exists now; `prod` is created later by applying the same Terraform root with `prod.tfvars` (bigger instance, longer backups). No staging: dev is the validation environment, and the load/stress test (step 5) runs against a temporary prod-sized stack. Rollback: previous image tag, plus database restore points (RDS snapshots before each prod migration).

7. Security and compliance posture

(Used later as the checklist for step 4.)

- **Boundaries:** RDS private only; EC2 security group allows 443/80 only; no SSH (SSM Session Manager); S3 public access blocked; CloudFront OAC.
- **IAM:** instance role limited to its S3 prefixes, SSM, ECR pull, CloudWatch; deploy role cannot change IAM; separate admin human roles with MFA.

- **App security:** HTTPS only, HSTS; JWT 15 min; refresh rotation with reuse detection; per-IP and per-user rate limits (Redis); CSRF protection for admin; strict CORS; Helmet; input validation; parameterised queries (Prisma); output escaping of question HTML via sanitiser; file upload type/size checks and AV-less pre-signed upload with content-type pinning.
- **Admin:** TOTP 2FA mandatory, recovery codes, audit log for every write, optional IP allow-list, read-only role.
- **Payments:** card data never touches our servers; webhook HMAC verification; idempotency keys; reconcile job against Razorpay settlement.
- **Test integrity:** server-side timing, session binding, tab-switch and copy event logging for attempts (log only; no proctoring promise).
- **Privacy / DPDP:** minimal PII (Google name/email/avatar, district); consent screen and privacy policy; data-export and delete-account endpoints (FR-006); logs scrub tokens; data stays in Mumbai; encryption in transit and at rest.
- **Backups / DR:** RDS automated backups with point-in-time recovery (14 d prod), weekly snapshot copy to a second region (prod), restore drill before launch. Target RPO ≤ 5 min, RTO ≤ 4 h.
- **Disclaimer** (NFR-010) shown in app, web footer and store listing.

8. Monthly cost estimate

Estimate dated 2026-10-27. Figures are from published on-demand AWS list prices as remembered at drafting time (Mumbai, USD) and converted at Rs 85/USD; they have not been checked against the live price list. Re-check in the AWS Pricing Calculator before approval. Razorpay fees (about 2% per transaction + GST) and domain/Play/EAS costs are excluded.

Service	Dev (now), USD	Prod (later), USD	Notes
EC2	6 (t4g.small, ~12 h/day)	28 (t4g.medium 24×7)	Reserved/Savings Plan can cut prod ~30%
EBS gp3 30 GB	3	3	
RDS PostgreSQL	13 (t4g.micro)	30 (t4g.small)	single-AZ, 20 GB gp3
RDS storage + backups	3	4	
Public IPv4 + Elastic IP	4	4	
S3 + CloudFront	2	6	~2 GB stored, ~100 GB/month transfer
Route 53 + ACM	1	1	
CloudWatch logs/alarms	3	6	log retention 14 days
Secrets/Parameter Store	1	2	
ECR	1	1	
Data transfer out (EC2)	1	4	
Weekly cross-region snapshot copy	–	2	
Total (USD)	**≈ 38**	**≈ 91**	
Total (INR @85)	**≈ Rs 3,200**	**≈ Rs 7,800**	

- Budget check: dev + prod together ≈ Rs 11,000, **within the Rs 5,000–15,000 target**; while only dev exists the bill is ≈ Rs 3,200.
- Headroom: ≈ Rs 4,000/month for a traffic spike (resize prod to t4g.large ≈ +Rs 2,300) or for Multi-AZ RDS (≈ +Rs 2,500).
- Optional extras if wanted (not in the numbers): Google Translation API for admin translations, an LLM for CA summaries, SMS later for phone login.

9. Risks, open questions, phases and effort

9.1 Risks

ID	Risk	Mitigation
R1	**Timeline:** Diwali 2026 is 8 Nov 2026, twelve days after this draft; the full MVP is ≈ 12–13 weeks for 2 people (§9.3).	Agree a cut-down "Diwali release" (below) or move the date. This is the biggest risk.
R2	Google Play may reject an app that sells digital content without Play Billing (user chose Razorpay only).	Prepare web purchase + direct APK; apply for Play policy review early.
R3	Single instance means 99.5% only, and a crash during a live test hurts.	Auto-recovery, autosave of attempts, test resumes within window; upgrade path to 2 instances behind ALB.
R4	Live tests and battle are realtime features that need load testing.	Load test in step 5 before the first live test.
R5	Redis in a container loses state on instance replacement.	Only rebuildable state; sessions are in Postgres, Redis is a cache of them.
R6	Current-affairs collection in the old admin used device-side browsing with the admin's own logins; a web admin cannot do this.	MVP: manual capsule writing and paste-URL extraction; automated collector deferred.
R7	PYQ and portal-sourced content may carry copyright.	Own-written summaries, link to sources (OQ-007).
R8	Content volume: someone must author questions and notes for 6 exams.	Bulk import, OCR later; this is a content, not an engineering, risk.

9.2 Open questions for the client

1. Is the target really **8 Nov 2026**, or Diwali 2027 (OQ-008)?
2. Accept Play-rejection risk with Razorpay-only (OQ-001a)?
3. Domain name and ownership; Google Cloud project for Sign-In; Razorpay account (KYC) status.
4. Who writes and uploads content before launch, and how many questions exist today?
5. Is a weekly cross-region backup copy required, or is Mumbai-only acceptable?
6. Should launch include offline downloads of notes (OQ-012)?

9.3 Delivery phases (two people, rough)

Phase	Scope	Effort
0. Foundations	repos, Terraform dev, CI/CD, auth + sessions, DB schema, i18n, design system	1.5 wk

Phase	Scope	Effort
1. Core content and tests	exams/topics/notes/PYQ, question bank + import, static tests, attempts, scoring, Razorpay plans and entitlements, admin basics	4 wk
2. Engagement	current affairs, analytics, leaderboard, doubts desk, planner, interview Qs, notifications	3 wk
3. Realtime	live tests, pre-test lobby, 1v1 battle with bots/replays	2.5 wk
4. Hardening	security review, load/soak test, DR drill, store release, prod stack	1.5 wk
Total		**≈ 12–13 weeks**

Diwali 2026 cut (≈ 4 weeks at best, not 12 days): only phases 0 and a thin phase 1 (Google login, one-device rule, free tests and notes for the 6 exams, current-affairs list, admin import). Payments, doubts, analytics, battle and live tests follow after. Even this does not fit before 8 Nov 2026, so the client must either move the date or accept a smaller, later first release.

10. Client sign-off

Approval checklist (client ticks each item):

- ☐ Scope and phases in §9.3 accepted, including what is deferred.
- ☐ Razorpay-only payments and Play-policy risk (R2) accepted.
- ☐ Google Sign-In only; phone number deferred.
- ☐ One-device-at-a-time rule (newest login wins) accepted.
- ☐ Dev first, prod later; no always-on staging.
- ☐ Monthly estimate (§8) and budget range accepted, subject to the AWS calculator re-check.
- ☐ 99.5% availability with single-AZ database accepted.
- ☐ Launch date decision recorded: _____

Client name: _____ Date: _____

Signature: _____

(`APPROVAL.md` was created on 2026-10-27 after the client signed v0.1.)

Amendments (after approval)

#	Date	Amendment	Reason	Flagged to
A1	2026-10-27	Terraform lives in its own repo, `exam-gadh-terraform` (modules, `envs/dev`, later `envs/prod`, infra CI workflows with OIDC). The backend repo no longer contains `infra/`. Architecture and cost are unchanged.	Project owner's instruction when approving the build layout	Project owner (in session); client to be told at next review

Repos are now: `exam-gadh-frontend`, `exam-gadh-backend`, `exam-gadh-terraform`, `exam-gadh-docs`. §6 and §5.1 references to `infra/` in the backend repo should be read as the Terraform repo.